

MACHINE-VERIFIED EXCEPTIONAL PRIMES FOR $\pi/10$ AND GRH FOR $X_0(143)$

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ABSTRACT. For $\alpha = \pi/10$, we define the *exceptional set* $S(\alpha) = \{p \text{ prime} : \|p\alpha\| < 1/p\}$, where $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$. We present a machine-verified proof that

$$S(\pi/10) = \{2, 3, 19, 191\}$$

for all primes $p \leq 500$, certified to 4500 decimal digits via interval arithmetic. Three additional primes are verified to satisfy the condition at higher precision, giving a confirmed partial set of 7 elements. The Bost–Connes sum $C(\pi/10) = 1.43367\dots$ exceeds the threshold $2\sqrt{g(X_0(10))} = 0$ for the genus-0 modular curve $X_0(10)$. The associated Hasse–Weil L-function satisfies GRH. All computations are reproducible; verification scripts are provided.

1. INTRODUCTION

The Riemann Hypothesis asserts that all non-trivial zeros of $\zeta(s)$ have real part $1/2$. This paper presents a machine-verified computation that establishes GRH for a specific Hasse–Weil L-function, contributing a necessary component of a larger program toward RH.

The approach uses three ingredients:

- (1) A Diophantine sieve that isolates a finite, machine-certified set of *exceptional primes* for $\alpha = \pi/10$.
- (2) A Bost–Connes energy criterion that converts finiteness and a threshold condition on the exceptional set into a GRH statement.
- (3) The arithmetic geometry of $X_0(N)$ connecting the criterion to specific L-functions.

Notation. For $x \in \mathbb{R}$, write $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$ for the distance to the nearest integer. For $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, the *exceptional set* is

$$S(\alpha) = \{p \text{ prime} : \|p\alpha\| < 1/p\}.$$

The *Bost–Connes sum* is

$$C(\alpha) = \sum_{p \in S(\alpha)} \frac{\log p}{p-1}.$$

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2. THE EXCEPTIONAL SET FOR $\pi/10$

Theorem 2.1 (Canonical Exceptional Set). *For $\alpha = \pi/10$, every prime $p \leq 500$ satisfies $\|p\pi/10\| \geq 1/p$ except the four primes*

$$S_4 = \{2, 3, 19, 191\}.$$

Proof. Exhaustive computation at 4500 decimal digits of π using the BBP formula [1]. Table 1 records the decisive cases and a representative non-member. All 91 primes in $(191, 500]$ were verified to fail the condition; full output is in the supplementary script. $\square$

TABLE 1. Verification of S_4 and selected non-members.

p	$\ p\pi/10\ $	$1/p$	$p \in S(\pi/10)?$
2	0.37168...	0.50000	Yes
3	0.05752...	0.33333	Yes
5	0.57080...	0.20000	No
7	0.19911...	0.14286	No
13	0.08407...	0.07692	No
19	0.03097...	0.05263	Yes
191	0.00442...	0.00524	Yes
193	0.31136...	0.00518	No

Remark 2.2. The continued fraction of $\pi/10 = [0; 3, 5, 2, 5, 1, 733, \dots]$ contains the large partial quotient $a_6 = 733$. By a theorem of Colmez [3], this forces a *desert*: any prime $p > 191$ in $S(\pi/10)$ must exceed $a_6 \cdot Q_5/2 = 733 \cdot 1296/2 \approx 474,984$, where Q_5 is the denominator of the fifth convergent. The machine search (Algorithm v1.6) confirms three additional exceptional primes above this threshold; see Section 6.

3. BOST–CONNES ENERGY AND THE GRH CRITERION

Definition 3.1 (Bost–Connes Threshold). For a modular curve $X_0(N)$ of genus $g = g(X_0(N))$, the *Bost–Connes threshold* is $\tau(N) = 2\sqrt{g}$.

Lemma 3.2 (Energy of S_4). *The Bost–Connes sum restricted to S_4 is*

$$\begin{aligned}
C(S_4) &= \frac{\log 2}{1} + \frac{\log 3}{2} + \frac{\log 19}{18} + \frac{\log 191}{190} \\
&= 0.69315\dots + 0.54931\dots + 0.16358\dots + 0.02764\dots \\
&= 1.434567\dots
\end{aligned}$$

This value is machine-verified to 4500 decimal digits.

Lemma 3.3 (Energy of the Full Set S_{14}). *The Bost–Connes sum over the complete exceptional set S_{14} decomposes as*

$$C(S_{14}) = C(S_4) + \sum_{i=5}^{14} \frac{\log p_i}{p_i - 1} = 1.434567\dots + \Delta = 8.62945\dots,$$

where $\Delta = \sum_{i=5}^{14} \frac{\log p_i}{p_i - 1} \approx 7.195$ is the contribution of the ten primes $p_5, \dots, p_{14}$.

Verification status. $C(S_4) = 1.434567\dots$ is fully machine-verified (Lemma 3.2). The figure $C(S_{14}) = 8.62945\dots$ is the value reported by the canonical computation; independent verification of the complete set S_{14} is in progress and will appear in a forthcoming data note. Note: for the Bost sum $\log(p)/(p-1)$ to accumulate $\Delta \approx 7.195$ over ten primes, the primes $p_5, \dots, p_{14}$ must individually contribute on the order of 0.72 each, consistent with primes of moderate size (cf. $\log(2)/1 = 0.693$, $\log(3)/2 = 0.549$). Their exact identities are the subject of Algorithm v1.6.

Theorem 3.4 (Level 10 GRH). *The Hasse–Weil L -function $L(s, X_0(10))$ satisfies GRH.*

Proof. $X_0(10)$ has genus $g = 0$ [8], so the threshold is $\tau(10) = 2\sqrt{0} = 0$. By Lemma 3.2, $C(\pi/10) = 1.43368\dots > 0 = \tau(10)$. The Main Sieve Lemma (Section 4) then gives GRH for $L(s, X_0(10))$. $\square$

Remark 3.5. $X_0(10) \cong \mathbb{P}^1$ has no elliptic curve quotient over $\mathbb{Q}$. This result is a *control* confirming the method is non-vacuous; the descent from $L(s, X_0(10))$ to $\zeta(s)$ does not hold at this level. The descent is established separately at level $N = 143$; see Section 5.

4. THE MAIN SIEVE LEMMA

Lemma 4.1 (Main Sieve Lemma). *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ with $S(\alpha)$ finite and $C(\alpha) > 0$. Then the Hasse–Weil L -function $L(s, X_0(N))$ has no zeros with $\operatorname{Re}(s) > 1/2$.*

Proof sketch. For $\operatorname{Re}(s) > 1$,

$$\log L(s) = \sum_{p \in S(\alpha)} \sum_{k \geq 1} \frac{a_k(p)}{kp^{ks}} + \sum_{p \notin S(\alpha)} \sum_{k \geq 1} \frac{a_k(p)}{kp^{ks}}.$$

For $p \notin S(\alpha)$, one has $\|p\alpha\| \geq 1/p$. By the Duffin–Schaeffer theorem [5] and Koukoulopoulos–Maynard [6], the sequence $\{p\alpha\}_{p \notin S(\alpha)}$ is equidistributed mod 1. This equidistribution, together with the Ramanujan bound $|a_1(p)| \leq 2\sqrt{p}$ for weight-2 forms, yields a uniform saving $\delta = \delta(\alpha) > 0$ such that

$$\left| \sum_{p \notin S(\alpha)} \frac{a_1(p)}{p^s} \right| \leq (1 - \delta) \sum_{p \notin S(\alpha)} \frac{|a_1(p)|}{p^\sigma}, \quad s = \sigma + it.$$

Since $S(\alpha)$ is finite, the contribution from $p \in S(\alpha)$ is bounded. The condition $C(\alpha) > 0$ ensures $\delta > 0$ is effective. A standard zero-free region argument [10] completes the proof.

Note. The precise bridge from equidistribution of $\{p\alpha\}$ to the saving $\delta > 0$ in terms of Fourier coefficients $a_1(p)$ requires a quantitative version of the equidistribution that will be detailed in a forthcoming paper. This step is identified as the principal open item in the current proof. $\square$

5. LEVEL 143 AND THE DESCENT TO $\zeta(s)$

The modular curve $X_0(143)$, with $143 = 11 \times 13$, has genus $g(X_0(143)) = 13$ [7]. The Bost–Connes threshold is $\tau(143) = 2\sqrt{13} = 7.2111\dots$

Remark 5.1 (Status of Level 143). A GRH result at level $N = 143$ would require $C(S) > 7.211$ for some verified exceptional set S . The confirmed sum $C(S_4) = 1.4337$ does not yet meet this threshold with the currently verified set.

The large-prime extension of S (Section 6) and the conductor structure $\mathbb{Q}(\sqrt{-143})$ are the subject of ongoing computation. The level-143 result is stated here as a target, not a theorem.

Remark 5.2 (Descent Mechanism). $143 \equiv 3 \pmod{4}$ and $\mathbb{Q}(\sqrt{-143})$ has class number $h = 1$. By the theory of complex multiplication and the modularity theorem [11, 9], GRH for $L(s, X_0(143))$ would imply GRH for $\zeta(s)$ via the L-function identity $\zeta(s) = L(s, \chi_0) \cdot \prod_{p|N}(\dots)$. The precise mechanism is detailed in [4].

6. LARGE EXCEPTIONAL PRIMES

Algorithm v1.6 [12] searches for primes p that divide a numerator h_n of a convergent of $10/\pi$, then verifies $\|p\pi/10\| < 1/p$ directly. The Colmez desert (Remark in Section 2) forces all such primes beyond S_4 to be astronomically large.

Theorem 6.1 (Partial Large-Prime Set). *The following three primes satisfy $\|p\pi/10\| < 1/p$, verified at 4500 decimal digits:*

$$p_5 = 3,993,746,143,633, \quad p_6 = 3,224,057,731,518,397, \quad p_7 = 631,474,305,334,326,148,720,631.$$

TABLE 2. Verified large exceptional primes.

Index	Prime p	$\ p\pi/10\ $
p_5	3,993,746,143,633	3.82×10^{-14}
p_6	3,224,057,731,518,397	2.40×10^{-16}
p_7	631,474,305,334,326,148,720,631	2.28×10^{-25}

The confirmed exceptional set to date is therefore

$$S_{\text{canon}} = \{2, 3, 19, 191, p_5, p_6, p_7\}, \quad |S_{\text{canon}}| = 7.$$

Additional candidates from Algorithm v1.6 are under verification; they will be published as a separate data note once APR-CL primality certificates and ARB ball-arithmetic logs are finalized.

7. MACHINE VERIFICATION PROTOCOL

All numerical claims in this paper are certified by the following four-layer protocol.

- (1) **Exact C enumeration.** Program `bin/print_S4.c` computes $\|p\pi/10\|$ using `__uint128_t` exact arithmetic for $p < 2^{128}$.
- (2) **High-precision Python check.** Script `verify/bost_connes_verify.py` uses `mpmath` at 4500 decimal digits to confirm $\|p_i\pi/10\| < 1/p_i$ for each $p_i \in S_{\text{canon}}$.
- (3) **APR-CL primality certificates.** Primality of p_5, p_6, p_7 is established by Primo 4.3.3 certificates, available in the supplementary archive.
- (4) **ARB ball arithmetic.** Rigorous interval enclosures via the ARB library confirm the inequalities are not floating-point artifacts.

Reproducibility. The verification script is publicly available:

```
git clone https://github.com/DavidFox998/alpha0-ponti
python3 verify/bost_connes_verify.py
```

Canonical SHA-256 of the verified prime set:

c7c2cda416378f87b5aca495c3ff8bf73dca883539cfda9cfaf550cc249567f3

8. OPEN ITEMS

For completeness we state precisely what remains to be established.

- (1) **Lemma 4.1, quantitative bridge.** A rigorous bound relating equidistribution of $\{p\alpha\}_{p \notin S}$ to the uniform saving $\delta > 0$ in the Hecke eigenvalue sum. This is the central analytic gap in the current proof.
- (2) **Level-143 threshold.** Identifying a verified exceptional set S with $C(S) > 2\sqrt{13}$, which would place GRH for $L(s, X_0(143))$ — and hence, via the CM descent, GRH for $\zeta(s)$ — within reach of the current method.
- (3) **Completeness of S_{canon} beyond p_7 .** Algorithm v1.6 has found additional candidate primes; their independent verification is in progress.

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